

Fourth Semester B. Tech. (Electronics and Communication Engineering / Electronics Design Technology) Examination

INTRODUCTION TO ELECTROMAGNETIC THEORY

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) Assume suitable data wherever necessary.
- (3) Given Constants : $\epsilon_0 = 8.854 \times 10^{-12}$ F/m and $\mu_0 = 4\pi \times 10^{-7}$ H/m.
- (4) **Bold letters represents vector.**

1. Solve any **Six** questions :

- (A) Transform vector $20\mathbf{a}_y$ into cylindrical coordinate system at point Q(x = 3, y = 4, z = -1). 2(CO1)
- (B) Write divergence operator for orthogonal curvilinear coordinate system and give it in cylindrical form with its significance. 2(CO1)
- (C) $D = \rho \cos\phi \mathbf{a}_\phi - (\sin\phi/3\rho) \mathbf{a}_z$ over the surface placed at Z=0 plane. ρ value is between 0 to a and $\phi = 0$ to $\pi/2$. Find the charge density using Gauss law. 2(CO2)
- (D) What is Poynting vector ? What does it signify ? 2(CO4)
- (E) Give properties of good conductor. 2(CO2)
- (F) A relation between **E** and **H** gives intrinsic impedance. Justify this statement. 2(CO4)
- (G) Given $\mathbf{E} = (x^2 + y^2)\mathbf{a}_x + (x + y)\mathbf{a}_y$, find Laplace operator (∇^2) of **E**. 2(CO3)
- (H) Show that at a boundary between two conducting media, the tangential component of electric field intensity is continuous. 2(CO3)

2. (A) Show that the work done in moving a point charge in an electric field is independent of the path. 3(CO2)
- (B) Given that $\mathbf{D} = 20\rho^2 \mathbf{a}_\rho$ C/m². The surface is bounded with $0 \leq \rho \leq 3$ and $0 \leq z \leq 2$. Evaluate either side of the Gauss Divergence theorem. 4(CO2)
- (C) Find the electric field at origin if the charge distribution present in free space are : point charge 12 nC at P(2, 0, 6); uniform line charge density 3 nC/m at $x = -2$, $y = 3$; uniform surface charge density 0.2 nC/m², at $x = 2$. 3(CO2)
3. (A) Find H in Cartesian component at P(2, 3, 4) if there is a current filament located at $x = -1$, $y = 2$, carrying current $I = 8$ mA. 4(CO3)
- (B) Point charge of +2 μ C and -2 μ C are located at (0, 0, 2) mm and (0, 0, -2) mm respectively in free space.
Find :
(i) Dipole moment p.
(ii) E at P(x = 1, y = 2, z = 1.5). 3(CO3)
- (C) Find the magnetic field at point Q(0, 2, 0) due to current element $I d\mathbf{L} = 2\pi \mathbf{a}_z \mu\text{A.m}$ located at P(4, 0, 0). 3(CO1,2)
4. (A) Evaluate either side of Stoke's Theorem,
Given : $\mathbf{H} = 6xy\mathbf{a}_x - 3y^2\mathbf{a}_y$ A/m for the rectangular loop is given, $2 \leq x \leq 5$, $-1 \leq y \leq 1$, $z = 0$ plane. 4(CO2,4)
- (B) Find the amplitude of the displacement current density in the air near an antenna, where the field strength of an FM signal is
 $E = 80 \cos (6.277 \times 10^8 t - 2.092y) \mathbf{a}_z$ V/m. 3(CO1,4)
- (C) State magnetic boundary conditions at an interface between two different magnetic materials. 3(CO1,2)

5. (A) The electric field intensity associated with a plane electromagnetic wave travelling in a perfectly dielectric medium is given by

$$E_x(z, t) = 10 \cos(3\pi \times 10^8 t - 2\pi z) \text{ V/m.}$$

Compute :

- (i) Frequency,
 - (ii) Wavelength,
 - (iii) Velocity of propagation,
 - (iv) Magnetic field intensity **H** if $\mu = \mu_0$. 4(CO3,4)
- (B) Given $E = E_m \sin(\omega t - \beta z) \mathbf{a}_y$ in free space. Find **D**, **B** and **H**. 3(CO3,4)
- (C) Explain in brief what is meant by skin effect and depth of penetration. 3(CO3)

6. Write short notes on any **Two** :—

- (i) Scalar magnetic potential.
- (ii) Significance of terms in power theorem (Poynting Theorem).
- (iii) Electric field at the boundary when it is incident normally on a plane conducting boundary.
- (iv) Gauss law and Divergence theorem. 8(CO2,3,4)

